

Topic 19

Capacitance

Revision Booklet

This booklet covers:

- Capacitors and Capacitance
- Capacitors in Series and Parallel
- Energy Stored in a Capacitor
- Capacitor Discharge
- The Time Constant

Core Concepts and Definitions

Capacitance

The **capacitance** of a conductor is the charge stored per unit potential difference across it.

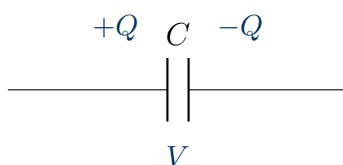
$$C = \frac{Q}{V} \quad \text{units: F (farads)} \equiv \text{C V}^{-1}$$

- 1 F is a very large unit; practical capacitors are typically μF , nF or pF .
- Capacitance depends on the **geometry** of the conductor and the medium between the plates, not on Q or V individually.
- For an **isolated spherical conductor** of radius R : $C = 4\pi\epsilon_0 R$.

The Parallel Plate Capacitor

Two conducting plates of area A separated by distance d :

- Charging the capacitor stores charge $+Q$ on one plate and $-Q$ on the other.
- The electric field between the plates is uniform: $E = V/d$.
- Capacitance increases with larger plate area A and smaller separation d .



Capacitors in Series and Parallel

Capacitors in Parallel

$$C_{\text{total}} = C_1 + C_2 + C_3 + \dots$$

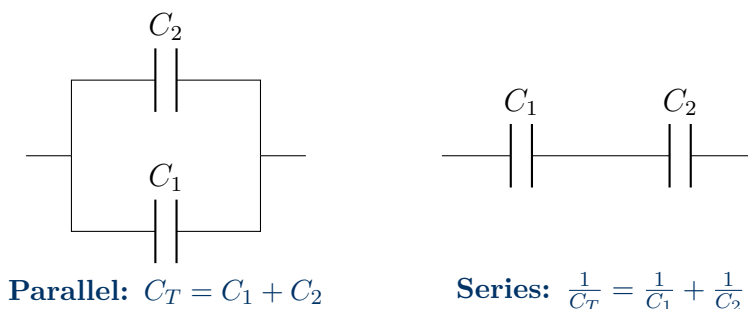
Each capacitor has the **same voltage**; charges add: $Q_{\text{total}} = Q_1 + Q_2 + \dots$

Capacitors in Series

$$\frac{1}{C_{\text{total}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$

Each capacitor has the **same charge**; voltages add: $V_{\text{total}} = V_1 + V_2 + \dots$

Circuit diagrams



Derivation of Series Formula

For capacitors in series, each carries the same charge Q . The total voltage is:

$$V_{\text{total}} = V_1 + V_2 = \frac{Q}{C_1} + \frac{Q}{C_2} = Q \left(\frac{1}{C_1} + \frac{1}{C_2} \right)$$

Since $C_{\text{total}} = Q/V_{\text{total}}$, dividing through by Q gives $\frac{1}{C_{\text{total}}} = \frac{1}{C_1} + \frac{1}{C_2}$.

Common Mistake

Capacitors in series combine like **resistors in parallel** (reciprocal rule), and capacitors in parallel combine like **resistors in series** (direct sum). This is the opposite to resistors — don't mix them up.

Energy Stored in a Capacitor

Energy Stored

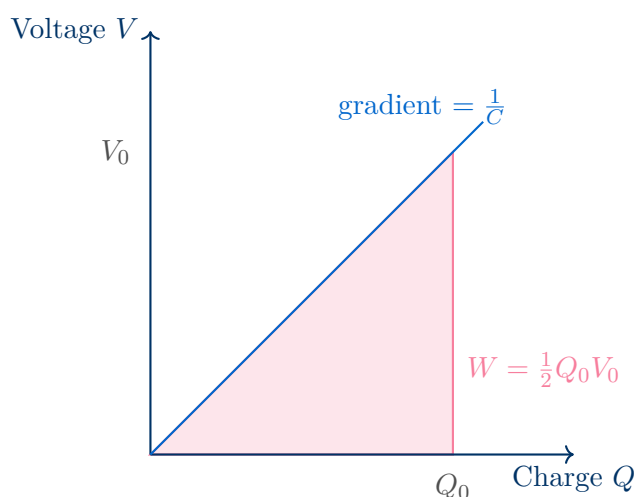
$$W = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$$

- W = energy stored (J)
- Q = charge stored (C)
- V = potential difference across capacitor (V)
- C = capacitance (F)

Energy from the V - Q Graph

The energy stored equals the **area under the V - Q graph** (a straight line through the origin with gradient $1/C$):

$$W = \text{area of triangle} = \frac{1}{2} \times Q \times V = \frac{1}{2}QV$$

V–Q graph for a capacitor**Discharging a Capacitor****Capacitor Discharge Through a Resistor**

When a charged capacitor discharges through a resistor R , the charge, voltage and current all decay **exponentially** with time:

$$Q = Q_0 e^{-t/RC} \quad V = V_0 e^{-t/RC} \quad I = I_0 e^{-t/RC}$$

where $I_0 = V_0/R = Q_0/RC$ is the initial current.

Time Constant

$$\tau = RC$$

τ = time constant (s)

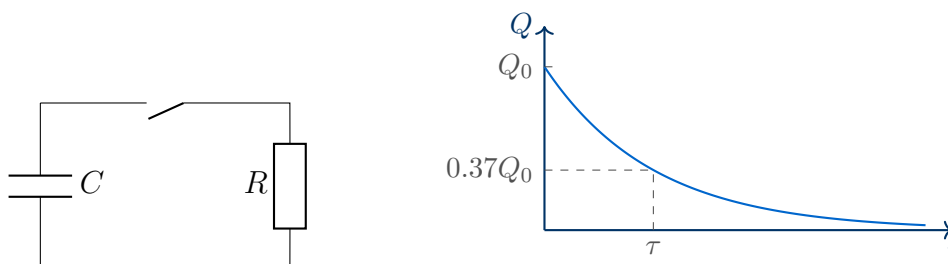
R = resistance of discharge path (Ω)

C = capacitance (F)

After one time constant ($t = \tau$), the charge/voltage/current has fallen to $e^{-1} \approx 37\%$ of its initial value.

Key Values During Discharge

- $t = \tau$: $Q = 0.37 Q_0$ (37%)
- $t = 2\tau$: $Q = 0.135 Q_0$ (13.5%)
- $t = 5\tau$: $Q \approx 0.007 Q_0$ — capacitor considered fully discharged.
- A larger τ means slower discharge; a smaller τ means faster discharge.

Discharge circuit and Q - t graph

Analysing Discharge Graphs

- A **linear** $\ln Q$ vs t graph confirms exponential decay; gradient $= -1/RC$.
- The time constant τ can be read directly as the time for Q to fall to $0.37Q_0$.
- Doubling R or C doubles τ and halves the rate of discharge.

Linearising the discharge: $\ln Q$ against t

Taking logarithms of $Q = Q_0 e^{-t/RC}$:

$$\ln Q = \ln Q_0 - \frac{1}{RC} t$$

This is of the form $y = mx + c$, so a graph of $\ln Q$ against t gives:

- gradient $= -\frac{1}{RC} = -\frac{1}{\tau}$
- y -intercept $= \ln Q_0$

Formula Summary Sheet

Formula	Quantity	Units
$C = Q/V$	Capacitance (definition)	F
$C_T = C_1 + C_2 + \dots$	Capacitors in parallel	F
$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$	Capacitors in series	F
$W = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$	Energy stored	J
$\tau = RC$	Time constant	s
$Q = Q_0 e^{-t/RC}$	Charge during discharge	C
$V = V_0 e^{-t/RC}$	Voltage during discharge	V
$I = I_0 e^{-t/RC}$	Current during discharge	A
$\ln Q = \ln Q_0 - \frac{t}{RC}$	Linearised discharge	—

Constants and values: $e^{-1} \approx 0.368$; after 1τ : 37% remains; after 5τ : < 1% remains

Units check: $[\tau] = [\Omega][F] = [V A^{-1}][C V^{-1}] = [C A^{-1}] = s$

Worked Examples

Example 1 — Capacitors in Series and Parallel

Question: Three capacitors of $2.0 \mu\text{F}$, $3.0 \mu\text{F}$ and $6.0 \mu\text{F}$ are connected (a) in parallel and (b) in series. Find the total capacitance in each case.

Solution

(a) Parallel:

$$C_T = 2.0 + 3.0 + 6.0 = \mathbf{11.0 \mu\text{F}}$$

(b) Series:

$$\frac{1}{C_T} = \frac{1}{2.0} + \frac{1}{3.0} + \frac{1}{6.0} = \frac{3 + 2 + 1}{6.0} = \frac{6}{6.0} = 1.0 \mu\text{F}^{-1}$$

$$C_T = \mathbf{1.0 \mu\text{F}}$$

Example 2 — Energy Stored

Question: A $470 \mu\text{F}$ capacitor is charged to 12 V. Calculate (a) the charge stored and (b) the energy stored.

Solution

(a) $Q = CV = 470 \times 10^{-6} \times 12 = \mathbf{5.64 \times 10^{-3} \text{ C}}$

(b) $W = \frac{1}{2}CV^2 = \frac{1}{2} \times 470 \times 10^{-6} \times 12^2 = \mathbf{3.38 \times 10^{-2} \text{ J}}$

Example 3 — Capacitor Discharge

Question: A $220 \mu\text{F}$ capacitor charged to 9.0 V discharges through a $47 \text{ k}\Omega$ resistor. Calculate (a) the time constant and (b) the voltage after 5.0 s .

Solution

(a) $\tau = RC = 47 \times 10^3 \times 220 \times 10^{-6} = \mathbf{10.3 \text{ s}}$

(b) $V = V_0 e^{-t/RC} = 9.0 \times e^{-5.0/10.3} = 9.0 \times e^{-0.485} = 9.0 \times 0.616 = \mathbf{5.5 \text{ V}}$

Practice Exam Questions**Section A — Short Answer Questions**

Q1. Define capacitance and state its SI unit.

[2 marks]

Q2. A capacitor stores $360 \mu\text{C}$ of charge when connected to a 12 V supply. Calculate its capacitance.

[2 marks]

Q3. Two capacitors of $4.0\ \mu\text{F}$ and $12\ \mu\text{F}$ are connected in series across a $6.0\ \text{V}$ supply.

(a) Calculate the total capacitance. *[2 marks]*

(b) Calculate the total charge stored. *[1 mark]*

(c) Calculate the voltage across the $4.0\ \mu\text{F}$ capacitor. *[2 marks]*

Q4. Explain, with reference to the V – Q graph, why the energy stored in a capacitor is $W = \frac{1}{2}QV$ and not $W = QV$.

[2 marks]

Q5. Define the time constant for a capacitor–resistor discharge circuit and state what fraction of the initial charge remains after two time constants.

[3 marks]

Section B — Longer Structured Questions

Q6. A $100\ \mu\text{F}$ capacitor is charged to $20\ \text{V}$ and then discharged through a $25\ \text{k}\Omega$ resistor.

- (a) Calculate the initial charge stored on the capacitor.

[1 mark]

- (b) Calculate the initial discharge current.

[2 marks]

- (c) Calculate the time constant for the discharge.

[1 mark]

- (d) Calculate the charge remaining after $4.0\ \text{s}$.

[2 marks]

- (e) The student plots a graph of $\ln(Q/C)$ against t/s . State the gradient and y -intercept of this graph.

[2 marks]

Q7. A $50\ \mu\text{F}$ and a $200\ \mu\text{F}$ capacitor are connected in series and charged from a $15\ \text{V}$ supply.

(a) Calculate the combined capacitance.

[2 marks]

(b) Calculate the energy stored in the combination.

[2 marks]

(c) The two capacitors are now reconnected in parallel across the same supply. Calculate the new total energy stored and explain why it differs from part (b).

[3 marks]

Mark Scheme and Answers

Q1. Capacitance is the charge stored per unit potential difference [1]; unit: farad (F) or C V^{-1} [1].

Q2. $C = Q/V = 360 \times 10^{-6}/12 = \mathbf{30\ \mu\text{F}}$ [2].

Q3(a). $\frac{1}{C_T} = \frac{1}{4.0} + \frac{1}{12} = \frac{3+1}{12} = \frac{4}{12}$ [1]; $C_T = \mathbf{3.0\ \mu\text{F}}$ [1].

Q3(b). $Q = C_TV = 3.0 \times 10^{-6} \times 6.0 = \mathbf{1.8 \times 10^{-5}\ \text{C}}$ [1].

Q3(c). Same charge on each capacitor in series; $V_1 = Q/C_1 = 1.8 \times 10^{-5}/(4.0 \times 10^{-6}) = \mathbf{4.5\ \text{V}}$ [2].

Q4. The V - Q graph is a straight line through the origin [1]; the energy is the area under this graph (a triangle), which is $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2}QV$ — not QV because the voltage builds up gradually from 0 to V as charge is stored [1].

Q5. The time constant is the product RC [1]; it is the time taken for the charge (or voltage or current) to fall to $1/e \approx 37\%$ of its initial value [1]; after 2τ : $e^{-2} \approx 13.5\%$ remains [1].

Q6(a). $Q_0 = CV_0 = 100 \times 10^{-6} \times 20 = 2.0 \times 10^{-3} \text{ C}$ [1].

Q6(b). $I_0 = V_0/R = 20/(25 \times 10^3) \text{ [1]} = 8.0 \times 10^{-4} \text{ A}$ [1].

Q6(c). $\tau = RC = 25 \times 10^3 \times 100 \times 10^{-6} = 2.5 \text{ s}$ [1].

Q6(d). $Q = Q_0 e^{-t/RC} = 2.0 \times 10^{-3} \times e^{-4.0/2.5} \text{ [1]} = 2.0 \times 10^{-3} \times e^{-1.6} = 2.0 \times 10^{-3} \times 0.202 = 4.0 \times 10^{-4} \text{ C}$ [1].

Q6(e). Gradient $= -1/RC = -1/2.5 = -0.40 \text{ s}^{-1}$ [1]; y -intercept $= \ln Q_0 = \ln(2.0 \times 10^{-3}) = -6.2$ [1].

Q7(a). $\frac{1}{C_T} = \frac{1}{50} + \frac{1}{200} = \frac{4+1}{200} = \frac{5}{200}$ [1]; $C_T = 40 \text{ }\mu\text{F}$ [1].

Q7(b). $W = \frac{1}{2}C_TV^2 = \frac{1}{2} \times 40 \times 10^{-6} \times 15^2 \text{ [1]} = 4.5 \times 10^{-3} \text{ J}$ [1].

Q7(c). $C_T = 50 + 200 = 250 \text{ }\mu\text{F}$; $W = \frac{1}{2} \times 250 \times 10^{-6} \times 15^2 \text{ [1]} = 2.8 \times 10^{-2} \text{ J}$ [1]; more energy is stored in parallel because the total capacitance is greater — more charge is drawn from the supply [1].

Revision Checklist

Use this checklist to track your understanding. Tick each box when you are confident:

Learning Objective	Confidence (1–3)
☐ Define capacitance; use $C = Q/V$	
☐ Calculate combined capacitance for series and parallel combinations	
☐ Derive the series and parallel formulae from $C = Q/V$	
☐ Use $W = \frac{1}{2}QV = \frac{1}{2}CV^2 = Q^2/2C$ for energy stored	
☐ Explain why energy stored is the area under a V – Q graph	
☐ Describe the exponential decay of Q , V and I during discharge	
☐ Use $x = x_0 e^{-t/RC}$ for discharge of charge, voltage or current	
☐ Define the time constant $\tau = RC$ and state its physical significance	
☐ Determine τ from a discharge graph (directly or via $\ln Q$ vs t)	
☐ Sketch discharge curves for Q , V and I against time	
☐ Linearise the discharge equation and interpret gradient and intercept	

Key: 1 = Need more work 2 = Getting there 3 = Confident

Good luck with your revision!

The exponential decay of a capacitor is one of the most important mathematical forms in physics. Once you can recognise it, linearise it, and extract τ from a graph, you have a powerful tool that reappears throughout the course.